

Module 2

Chapter 6

Linear Regression: Line of Best Fit

Suppose we have acquired lots of data with multiple variables. How do we make any sense of it?

One question we can ask is: How are they related? For example, we could ask for the relationship between people's weights and heights, or study time and test scores, or two animal populations. **Regression** is a set of techniques for estimating such relationships and the simplest type of relationship is **Linear**. This process is unsurprisingly called **linear regression**, and it has many applications.

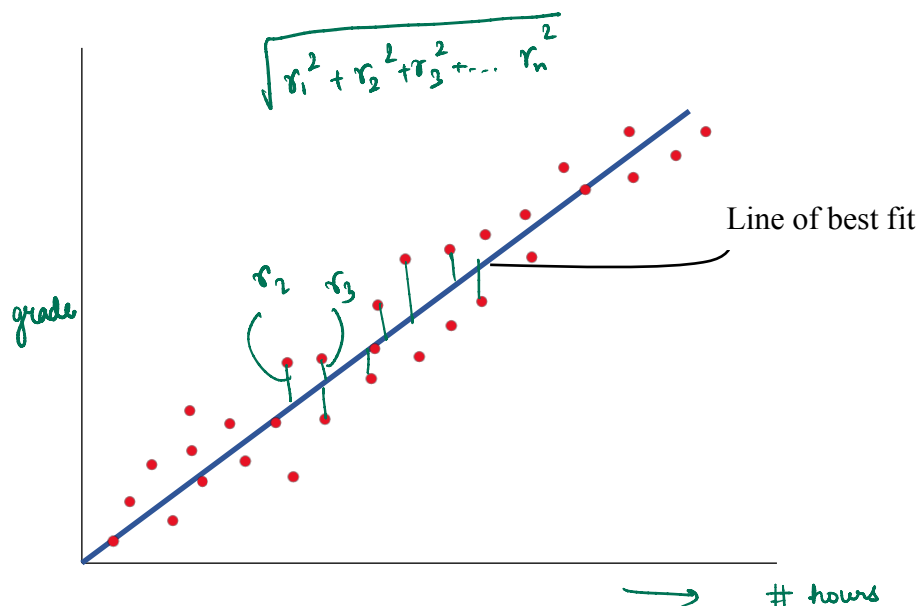


Fig 1

Simple linear Regression

Assumptions

Before we begin unraveling the world of linear regression, let us first talk about the assumptions that are made in linear regression that ensures reliability of the model results:

- **Linearity:** It assumes that there is a linear relationship between the independent and dependent variables. This means that *changes in the independent variable lead to proportional changes in the dependent variable.*
- **Independence:** The observations should be independent from each other that is the *errors from one observation should not influence other.*
- **Homogeneity of Variance:** the *size of the error* in our prediction *doesn't change significantly* across the values of the independent variable.
- **Normality:** the data follows a *normal distribution.*

Line of best Fit

Our primary objective while using linear regression is to locate the **best-fit line**, which implies that the **error** between the **predicted** and **actual** values should be **minimum**. There will be the least error in the best-fit line. The best Fit Line equation provides a **straight line** that represents the relationship between the dependent and independent variables. (Fig 1)

We're going to fit a line $y = mx + b$ to our data. Here, x is called the **independent variable or predictor variable**, and y is called the **dependent variable or response variable**. (Linear regression performs the task to *predict* a dependent variable value (y) based on a given independent variable (x)). Hence, the name is Linear Regression.

Before we talk about how to do the fit, let's take a closer look at the important quantities from the fit:

- **m** is the slope of the line: this is one of the most important quantities in any linear regression analysis. The slope of the line indicates how much the dependent variable changes for a unit change in the independent variable(s).
- **b** is the intercept of the line; The value of y when x is 0

In order to actually fit a line, we'll start with a way to quantify how good a line is. We'll then use this to fit the "best" line we can.

How to find line of best fit?

The idea of ***projection*** that we learnt in the last chapter can be used to fit a line to data in a way that each data point is *as close as possible* to the line of best fit. We can understand fitting a curve generally as a projection from the space that the data lives in originally, to the space of possible lines that could fit that data. The method we'll talk about is also known as a "***least-squares fit***" because it ***minimizes the sum of the squared error of each data point, relative to the fit's prediction***.

We aim to try to find the ***closest line*** to the data points that we have. The point that is receiving the projection consists of all our data wrapped up into one point. As for the range of the projection, we're actually going to look at the space of possible lines, as defined by the parameters of slope and y-intercept, and project onto that. For lines of the form **$y = mx + b$** , we actually have two independent parameters, **m** and **b** , and we'll want to find the best line that lives in that two-dimensional space.

Let's work with an example:

Case 1: Line of best fit with 0-intercept ($y = mx$)

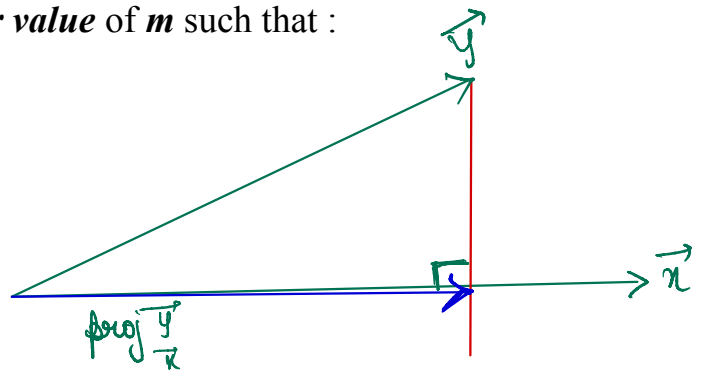
We'll start by modeling a situation where we don't need a y-intercept. Suppose we have run three experiments of flipping a coin, and we've received the following results:

- *30 flips: 16 heads*
- *60 flips: 34 heads*
- *90 flips: 51 heads*

We'd like to make a hypothesis that given x flips, the number of heads to expect is $y = mx$, where m is some constant describing the probability of heads on the coin. So, one way to look at this is that we have 3 data points of the form (x, y) , and we're trying to fit a line to them.

In an ideal world, there would be a *singular value* of m such that :

$$\begin{cases} 16 = m \cdot 30 \\ 34 = m \cdot 60 \\ 51 = m \cdot 90 \end{cases}$$



In vector form, this can be written as:

$$\underbrace{\begin{bmatrix} 16 \\ 34 \\ 51 \end{bmatrix}}_{\vec{y}} = m \underbrace{\begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}}_{\vec{x}}$$

But, real world data is noisy, and there's no such number that satisfies all three observations. Instead, we have a *space of possible predictions* :

$$m \begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}$$

and we want to choose the m that results in a prediction that *best fits* or *closest* to the data (y 's) we've already observed.

The space of possible predictions given m is a line in the direction of $\begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}$ and

we want the one that is closest to the point $\begin{bmatrix} 16 \\ 34 \\ 51 \end{bmatrix}$

Since this is just a line we're projecting onto, we can use our *projection onto a line equation*. Let's see how:

$$Proj_{\vec{x}} \vec{y} = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} \vec{x} = m \vec{x} \quad m = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}}$$

Here, $\vec{x}$ is $\begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}$, $\vec{y}$ is $\begin{bmatrix} 16 \\ 34 \\ 51 \end{bmatrix}$ and the projection of our data onto the line is given by the above equation.

Since we only care about our parameter m , we can just solve for that coefficient:

$$m = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}} = \frac{\begin{bmatrix} 16 \\ 34 \\ 51 \end{bmatrix} \cdot \begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}}{\begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix} \cdot \begin{bmatrix} 30 \\ 60 \\ 90 \end{bmatrix}} = 0.56$$

So, our equation of the line of best fit becomes

$$y = 0.56x$$

Let's Practice

Suppose we have gathered the following data for the mile checkpoints of an average athlete during their marathon training:

Time $t = 7$ (mins), mile = 1

Time $t = 13$ (mins), mile = 2

Time $t = 22$ (mins), mile = 3

Time $t = 27$ (mins), mile = 4

We want to find the line of best fit of the form $y = mx$ for the above data. What will be the equation of that line?

*you want to predict time
taken for a given
number of miles*

Case 2: line of best fit with a non-zero intercept ($y = mx + b$)

We are essentially projecting onto a two-dimensional space, since both m and b can vary.

Let's take an example

Let us suppose, we have been given the following data points:

(0,0), (1,2), (2,7), (3,10)

And we want to find the line of best fit of the form $y = mx + b$ for the above data.

In an ideal world, we will have singular values of m and b such that:

$$0 = m \cdot 0 + b$$

$$\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

$$\vec{y} = \begin{bmatrix} 7 \\ 13 \\ 22 \\ 27 \end{bmatrix}$$

line of best fit: $y = mx$

$$m = \frac{\vec{y} \cdot \vec{x}}{\vec{x} \cdot \vec{x}}$$

$$= \frac{\begin{bmatrix} 7 \\ 13 \\ 22 \\ 27 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}}{\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}}$$

$$\frac{7 + 26 + 66 + 108}{1 + 4 + 9 + 16}$$

$$= \frac{207}{30} = 6.9$$

$\therefore y = 6.9x$ is our equation for

the line of best fit

$$2 = m * 1 + b$$

$$7 = m * 2 + b$$

$$10 = m * 3 + b$$

Writing this in matrix notation:

$$\underbrace{\begin{bmatrix} 0 \\ 2 \\ 7 \\ 10 \end{bmatrix}}_{\vec{y}} = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} m \\ b \end{bmatrix}}_{\vec{c}}$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix}$$

$$y = mx + b$$

$$\text{Proj}_{\vec{m}} \vec{y} = A \vec{c}$$

$$\vec{c} = \underline{(A^T A)^{-1} A^T \vec{y}}$$

Unfortunately, we cannot find such values of m and b which satisfy all the above equations.

What do we have now?

On the Right Hand Side, we have a matrix A that describes what we know about each of the data points. Note that there is a column of 1's that represents a part of the results that that doesn't depend on the data (corresponding to the b term).

*Since we couldn't find singular values of m and b which satisfy all the above equations, we will now find those values of m and b which when multiplied to matrix A , give us **predicted** values of y as close to the **original** y as possible*

So, we find the closest point in the space of possible predictions $A\vec{c}$ that corresponds to the actual data observed - here again represented with a single point of very high dimension.

Remember the equation for projecting onto spaces that aren't lines?

$$\mathbf{Proj}_M \vec{y} = A\vec{c}$$

We want to find the values of $\mathbf{m}$ and $\mathbf{b}$. So, we are interested in finding the vector $\vec{c}$:

$$\vec{c} = (A^T A)^{-1} A^T \vec{y}$$

Here,

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$$

$$(A^T A)^{-1} A^T \vec{y} = \left(\begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 0 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 7 \\ 10 \end{bmatrix}$$

Thus,

$$\vec{c} = \begin{bmatrix} 3.5 \\ -0.5 \end{bmatrix}$$

Our line of best fit for our data now becomes:

$$y = 3.5x - 0.5$$

A Question that might arise here is: How does this approach guarantee to minimize the sum of squares error?

Well, we've created a space where each dimension corresponds to a different data point. And we've chosen a prediction, consisting of predicted values, that is as *close as possible* in this space to the original data - we know that because the ***angle between the subspace and the vector to the original data is orthogonal.***

The distance between the vector of original data and the vector of predictions is

$$\sqrt{r_1^2 + r_2^2 + \dots + r_n^2}$$

where each r is the ***residual*** for that data point. Since minimizing this length also minimizes its square, we've ***minimized the sum of the squared residuals for the prediction.***

Let's Practice

Find the equation of the line of best fit of the form $y = mx + b$ for the following data:
(-1,2) , (0,0), (1,-3), (2, -5)

More linear fits

There is data everywhere and not all data may be linear. How do we deal with them using the knowledge we have accumulated till now?

Step 1: Construct vector $\vec{y}$ and $\vec{x}$

$$\vec{y} = \begin{bmatrix} 2 \\ 0 \\ -3 \\ -5 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

Step 2: Construct Matrix A

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}$$

Step 3: Construct vector $\vec{c}$

$$\vec{c} = \begin{bmatrix} m \\ b \end{bmatrix}$$

Step 4: To find m and b , we find $\vec{c}$

$$\vec{c} = (A^T A)^{-1} A^T \vec{y}$$

step 5: $(A^T A) = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}$

$2 \times 4 \qquad 4 \times 2 \qquad 2 \times 2$

step 6: $(A^T A)^{-1} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} 1/5 & -1/10 \\ -1/10 & 3/10 \end{bmatrix}$

step 7: $(A^T A)^{-1} A^T = \begin{bmatrix} 1/5 & -1/10 \\ -1/10 & 3/10 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

$2 \times 2 \qquad 2 \times 4$

$= \begin{bmatrix} -3/10 & -1/10 & 1/10 & 3/10 \\ 4/10 & 3/10 & 2/10 & 1/10 \end{bmatrix}$

2×4

step 8: $(A^T A)^{-1} A^T \vec{y} = \begin{bmatrix} -3/10 & -1/10 & 1/10 & 3/10 \\ 4/10 & 3/10 & 2/10 & 1/10 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -3 \\ -5 \end{bmatrix}$

$2 \times 4 \qquad 4 \times 1$

$= \begin{bmatrix} -24/10 \\ -3/10 \end{bmatrix} = \begin{bmatrix} -12/5 \\ -3/10 \end{bmatrix}$

$2 \times 1 \qquad 2 \times 1$

$$\Rightarrow \vec{c} = \begin{bmatrix} -12/5 \\ -3/10 \end{bmatrix} = \begin{bmatrix} m \\ b \end{bmatrix}$$

$$\therefore m = -12/5$$

$$b = -3/10$$

$\therefore$ Our line of best fit is

$$y = -\frac{12}{5}x - \frac{3}{10}$$

Fortunately, linear models can be more versatile than they appear – even if the output isn't linear in the input $\vec{x}$, it could still be linear in some function $f(\vec{x})$ that is easily computed from x .

Case 1: Exponential models

Let us have an exponential growth model of the form : $y = e^{mx}$ (this is something that can commonly be encountered in economics or models of human behavior).

We can see that the function isn't particularly linear itself, but can fit the model into *natural logs of y*.

Now, $y = e^{mx}$ becomes:

$$\ln y = \ln(e^{mx}) = mx \quad \frac{\ln y}{y} = mx$$

Though y isn't *linear* in x , $\ln y$ is. We can then perform a least squares fit as if this were just another linear model.

Let's take an example

Suppose we want to fit a model of the form $y = e^{mx}$ to the three points :

(1, 5), (2, 8), and (3, 11).

Taking the *logs* of the three *y-coordinates* gives new points to fit:

(1, 1.6), (2, 2.1), and (3, 2.4).

Fitting $\ln y = mx$ is just projecting the vector:

$$\overrightarrow{\ln y} = \begin{bmatrix} 1.6 \\ 2.1 \\ 2.4 \end{bmatrix}$$

onto the span of $\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

This is a one-dimensional projection, so our formula for the coefficient m is :

$$m = \frac{\overbrace{\ln y}^y \cdot \vec{x}}{\vec{x} \cdot \vec{x}} = \frac{\begin{bmatrix} 1.6 \\ 2.1 \\ 2.4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}}{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}} = \frac{13}{14}$$

So, the equation of our line is

$$y = e^{\left(\frac{13}{14}\right)x}$$

Let us compare the original y 's to the predicted y 's as per our line of best fit:

Original y 's	Predicted y 's as per our line of best fit
5	$e^{\left(\frac{13}{14}\right)*1} = 2.5$
8	$e^{\left(\frac{13}{14}\right)*2} = 6.4$
11	$e^{\left(\frac{13}{14}\right)*3} = 16.3$

The **error** gets **bigger** in absolute terms for larger inputs. Why do you think this is the case?

---- When we changed the units to the **log of y** and then tried to find the best fit, we were actually **minimizing the sum of squared residuals** in that transformed **log space**. Equal errors in the log at different magnitudes result in a **larger absolute error** when the underlying magnitude is large. (We see this kind of effect with the Richter scale for earthquakes, where the leap from a 7.5 to an 8.0 is much more devastating than a change from a 2.5 to a 3.) So this is a downside of transforming nonlinear functions into linear ones to find a least squares fit; the error can behave in unexpected ways, or value data points unequally because of the transformation.

How can we achieve a better fit?

We can achieve a better fit with a more general exponential model of the form

$$y = Ce^{mx}$$

$$\ln C = b$$

$$\vec{c} = \begin{bmatrix} m \\ \frac{\ln C}{b} \end{bmatrix}$$

The constant C allows a change of **y-intercept**, which would otherwise have to be 1 for e^0 . Taking the **natural log of both** sides results in the equation:

$$\ln y = \ln C + mx$$

$$y = mx + b$$

$$\text{where } b = \ln C$$

If we treat $\ln C$ as b and $\ln y$ as our **new y** value, this again is a fit of the form

$$\vec{c} = \begin{bmatrix} m \\ b \end{bmatrix}$$

$$y = mx + b$$

$$b = \ln C$$

We can solve for m and b with a projection, then compute $C = e^b$ to get back the other parameter of the exponential function. Doing this with our example points above gives:

$m = 0.4$, $b = 1.23$, so

$$C = e^{1.23} = 3.43$$

The model now becomes:

$$y = 3.43e^{0.4x}$$

and our three model predictions become a much more reasonable:

$(1, 5.12)$, $(2, 7.64)$, and $(3, 11.40)$.

So, arguably the previous model just needed another degree of freedom to move its intercept.

Case 2: Multiple Regression

$$\vec{c} = \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_n \\ b \end{bmatrix} \quad A = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \vec{x}_3 & \cdots & \vec{x}_n \\ 1 \end{bmatrix}$$

Sometimes we may want to use *multiple predictor variables* to predict a given value, so

$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b$$

For example, housing prices could be predicted by a few variables: number of bedrooms, number of bathrooms, the year it was last renovated, and so on. If we think we can phrase the contribution of each factor as contributing to the final outcome at a fixed rate, like “one thousand dollars per bathroom,” then a linear fit is a good choice.

The reasoning and the methods are identical, except now the *matrix A* has *one column per predictor variable*. We still also want the *all-ones column*, assuming there is a fixed term b in the final equation. The process of finding the coefficients m_i is the same, computing

$$\vec{c} = (A^T A)^{-1} A^T \vec{y}$$

where $\vec{y}$ is the *dependent variable* vector. The *coefficients* in $\vec{c}$ follow the *same order as the columns of A*.

An *issue* that is new to multiple regression is that it is somewhat dangerous to include *predictors* that are *highly correlated* with each other; the algorithm may not know which predictor to blame. For example, if we wanted to predict height with left leg length and right leg length, the algorithm could choose coefficients that place the blame entirely on the left leg, with the right leg coefficient 0, or it could do the opposite and make a model that predicts on the basis of the right leg alone. Either of these models could work perfectly well, and the algorithm might declare the other variable unnecessary with a zero coefficient. In practice, the algorithm will probably share the blame and give a nonzero coefficient to each, but the relative sizes of those coefficients could still be *misleading*.

Case 3: Fitting Polynomials

Linear regression can actually be rather versatile, fitting nonlinear models such as *polynomials*. Polynomials definitely are themselves *nonlinear*, as *squares* and *cubes* render a function nonlinear. One way to fit polynomial models is to trick the algorithm to treat it as a basic multiple regression. Finding the best fit here still minimizes the sum of squared error, just as in basic multiple regression. We just “trick” the algorithm into treating *powers* of one predictor variable as if they were entirely *new observations*:

So, if we want to fit our data to a polynomial of the form:

$$y = m_1 x^1 + m_2 x^2 + \dots + m_n x^n + b$$

$$y = m_1 x_1 + m_2 x_2 + \dots + m_n x_n + b$$

$$A = \begin{bmatrix} x_1 & x_2 & x_3 & \dots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} m_1 \\ m_2 \\ \vdots \\ m_n \\ b \end{bmatrix}$$

We can treat each power of the predictor variable, x , as a separate predictor variable, thereby making it similar to multiple regression:

$$y = m_1x_1 + m_2x_2 + \cdots + m_nx_n + b$$

Where

$$x_1 = x^1$$

$$x_2 = x^2$$

.

.

.

$$x_n = x^n$$

Let's take an example:

Let our observations be:

$(-2, 8), (-1, 3), (0,0), (1, 2), (2, 5)$.

We want to fit a parabola to this data:

$$y = m_2x^2 + m_1x + b$$

Our basis matrix A contains a column for x^2 , a column for x , and a column of ***all ones*** for the offset (b). So, A becomes:

$$A = \begin{bmatrix} 4 & -2 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

Computing $(A^T A)^{-1} A^T \vec{y}$ gives *coefficients* $(1.5, -0.7, 0.6)$

↓ ↓ ↓
 $m_2 \ m_1 \ b$

Thus, our polynomial becomes:

$$\underline{\underline{y = 1.5x^2 - 0.7x + 0.6}}$$

Let's Practice

Here is the data collected for rent trends in Boston in 2020:

Rent (\$)	Number of bedrooms	Number of washrooms	Carpet area
3000	1	1	450
3600	2	1	600
5500	3	2	900

We are trying to fit the data into the following line in order to predict the rent:

$$y = m_1 x_1 + m_2 x_2 + m_3 x_3 + b$$

Find the basis matrix A for the above data

$$A = \begin{bmatrix} 1 & 1 & 450 & 1 \\ 2 & 1 & 600 & 1 \\ 3 & 2 & 900 & 1 \end{bmatrix}$$

$$\vec{c} = \begin{bmatrix} m_1 \\ m_2 \\ m_3 \\ b \end{bmatrix}$$

(we are trying to predict the rent. So,
rent is our $\vec{y}$ vector)

Extra Practice :

1. Find the Rank of the given matrix. Is the Matrix singular? Use Elementary Row Operations

$$\begin{bmatrix} -1 & 2 & -2 \\ 4 & -3 & 4 \\ -2 & 4 & -4 \end{bmatrix}$$

4x5

What is the maximum possible Rank of this Matrix?
What is the determinant of the matrix?

2. Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & 1 \\ 5 & 3 & 6 \end{bmatrix}$

Find the minor of a_{22}

What is the co-factor of a_{22} ?

3. Find the co-factor matrix of:

$$\begin{bmatrix} 3 & 0 & 4 \\ 2 & -1 & 3 \\ -3 & 5 & 1 \end{bmatrix}$$

$$1) \rightarrow A = \begin{bmatrix} -1 & 2 & -2 \\ 4 & -3 & 4 \\ -2 & 4 & -4 \end{bmatrix}$$

3x3

$$\therefore \rho(A) \leq 3$$

$$\downarrow R_3 = R_3 - 2R_1$$

$$\begin{bmatrix} -1 & 2 & -2 \\ 4 & -3 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\downarrow R_2 = R_2 + 4R_1$$

$$\begin{bmatrix} -1 & 2 & -2 \\ 0 & 5 & -4 \\ 0 & 0 & 0 \end{bmatrix}$$

→ Echelon Form

(• Every Non zero Row has a pivot
• Pivot of a row is to the right of the pivot of the row above it)

$$\therefore \text{Rank} = 2$$

Matrix is singular

$$\text{Determinant} = 0$$

$$2) \quad a_{22} = 0$$

$$\therefore \text{Minor} \rightarrow \begin{vmatrix} 1 & 3 \\ 5 & 6 \end{vmatrix}$$

$$= 6 - 15$$

$$= -9$$

$$\begin{aligned} \text{Co factor of } a_{22} &= (-1)^{2+2} \times -9 \\ &= -9 \end{aligned}$$

$$3. \quad \text{Co-factor Matrix: } \begin{bmatrix} -16 & -11 & 7 \\ 20 & 15 & -15 \\ 4 & -1 & -3 \end{bmatrix}$$

$$\text{Eg. Co-factor of } 3 : (-1)^{1+1} \times \begin{vmatrix} -1 & 3 \\ 5 & 1 \end{vmatrix} = -16$$

$$\text{Co factor of } 0 : (-1)^{1+2} \times \begin{vmatrix} 2 & 3 \\ -3 & 1 \end{vmatrix} = -11$$

and so on